

SEMANTIC WEB PROJECT

VN-Edu LOD

A Knowledge Graph of Vietnamese Higher-Education Institutions

Design, Implementation, and Critical Evaluation

Revised English technical report

6 October 2026

This edition incorporates an independent code and data audit of repository revision 21060d4. Measurements are recomputed from the corrected release rather than copied from the earlier Vietnamese report. The earlier report remains an archival description; its accuracy, security, and performance claims are not evidence for this edition.

Project repository:

<https://github.com/pham-ng/Vietnam-University-Knowledge-Graph-ver2>

Abstract

VN-Edu LOD integrates information about Vietnamese educational institutions into an RDF knowledge graph, combining structured Wikidata records, Vietnamese Wikipedia infoboxes, and curated reference tables. The corrected snapshot contains 300 educational-organization records, of which 271 are classified as higher-education institutions under the project's classification rules. It also represents 1,702 person records, 34 current and 29 former province-level units, and 78 illustrative academic programs. An ontology supports organizational, geographic, and educational relations; rule-based materialization adds inverse relations, classifications, and geographic containment. JSON Schema and SHACL enforce selected data contracts.

The resulting serving graph contains 48,492 distinct triples. An independent review identified defects in SPARQL request validation, query cancellation, publication gates, entity identity, persistent URI allocation, and the interpretation of evaluation results. The revised implementation addresses these defects and adds targeted regression tests. Evaluation distinguishes local structural validity, reference-link agreement, and factual accuracy. No independent population-level accuracy estimate, formal OWL-profile certificate, or current public-service availability measurement is claimed. The result is a reproducible educational Linked Data system with explicit operational and epistemic limits.

Keywords: knowledge graph; RDF; ontology; SHACL; SPARQL; entity resolution; data provenance; Vietnamese higher education.

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1 Introduction and scope

Information about a university can appear under different names and identifiers across encyclopedic articles, knowledge bases, and organizational websites. Relationships are equally important: a member school is different from a branch campus; direct governance is different from general state oversight; and the area containing an institution may have changed its administrative representation. A knowledge graph makes these distinctions explicit and supports queries that join several kinds of entity.

This project is an engineering case study of that integration problem. Its contribution is the combination of a domain model, a layered data pipeline, machine-readable outputs, and a query interface. It does not introduce a new reasoning algorithm or establish a complete official register of Vietnamese higher education. Names, classifications, and dates reflect the collected snapshot and documented extraction policies.

The evaluation is organized around four questions:

1. Can the model distinguish relevant organizational relationships and support useful multi-step queries without introducing contradictions detected by the implemented checks?
2. Can the pipeline reproduce a validated release and preserve stable identifiers as labels and input ordering change?
3. What do the available data and evaluation procedures establish about completeness, link agreement, and correctness, and what remains unmeasured?
4. Can publication and public query handling fail safely on invalid data and hostile inputs?

The audit covers source code, the ontology, SHACL shapes, cached inputs, generated RDF, tests, and the existing report. Security counterexamples are executed locally; external production endpoints are not used as attack targets. The project author's identity, institution, course identifier, and supervisor are not inferred or fabricated.

2 Technical foundations

RDF describes resources through triples. Subjects may be IRIs or blank nodes; predicates are IRIs, and objects may additionally be literals. A graph is a set, so duplicate triples do not increase its size. This matters when combining asserted data, ontology statements, links, and inferred statements: their file counts cannot simply be added to obtain the size of the union [1].

OWL axioms express implications about the modeled world. A property's domain is not an input-validation rule: using the property can entail that its subject belongs to that domain. Similarly, a functional object property can imply equality between two values; it does not necessarily reject the second value. The project therefore supplements reasoning with explicit validation of asserted records. The OWL 2 profile specification defines restrictions under which particular reasoning guarantees apply [2].

SHACL validates a supplied data graph against declared constraints [3]. Conformance is relative to those constraints, not a claim that every statement is true. Here, missing values for some descriptive fields generate warnings, while invalid datatypes, selected cardinality violations, and hard constraints prevent publication.

SPARQL supplies query syntax and algebra for RDF, including optional patterns, aggregates, and remote SERVICE clauses [4]. Exposing it over HTTP introduces an execution boundary: query syntax, remote destinations, elapsed work, and publication state must all be controlled independently of the user interface.

The project reuses identifiers and vocabulary from Wikidata, DBpedia, schema.org, FOAF, SKOS, and PROV-O. PROV-O supports descriptions of derivation [5]; whether the provenance is attached to an entity, a graph, or an individual assertion determines what can actually be traced. HTTP identifiers and useful representations support Linked Data publication, while persistence requires more than generating readable slugs [6].

3 Data sources and collection

3.1 Source roles

Wikidata contributes identifiers, labels, institutional attributes, organizational links, and person-related records. Vietnamese Wikipedia contributes infobox fields, descriptions, historical text, and links between articles. DBpedia supplies candidate identity links and some fallback values. Selected reference tables describe province mergers, regional groupings, fields of study, majors, and example programs. Geocoding is used as a fallback for location resolution and is not equivalent to an authoritative campus register.

Wikipedia and Wikidata are interconnected community resources. Agreement between them is useful corroborating evidence, but they must not be described as statistically independent sources. Imported statements can share an origin, and one resource may repeat an error from another. A disagreement log exposes selected conflicts; it does not establish which source is correct.

The collection strategy combines article-category traversal, infobox parsing, and Wikidata queries. Results are retained in JSON snapshots. The repository also ships a compressed HTTP cache. In the revised default pipeline, a cache miss raises an error instead of silently contacting a changing remote source. A fresh collection is an explicit operation. Dependency installation may still require Internet access; the offline claim applies to cached data collection and reconstruction after dependencies are installed.

3.2 Scope and selection effects

The snapshot is a discovered set of educational organizations, not a census. Filtering removes unrelated articles and some records lacking sufficient cross-source evidence. That policy can exclude legitimate but poorly documented institutions. Institutions with more detailed encyclopedia coverage are more likely to have richer records. The same bias is particularly strong for people linked through education records.

The retained higher-education classification includes name-based heuristics. A title beginning with a familiar institution type is evidence used by the implementation; it is not a substitute for a current legal decision. Counts of higher-education institutions must therefore be read as counts under the project's classification policy.

Academic programs are illustrative hand-entered data. They demonstrate program-to-major queries but do not support claims about a university's complete or current admissions offerings. Similarly, a Wikidata education relation establishes recorded attendance or education, not necessarily completion of a degree. The existing *Alumnus* class must be interpreted with that limitation.

4 Architecture and data management

4.1 Layers and contracts

Bronze stores responses and extraction snapshots. Silver contains institutions, people, governing bodies, and provinces in a normalized JSON representation. Gold contains asserted RDF, external links,

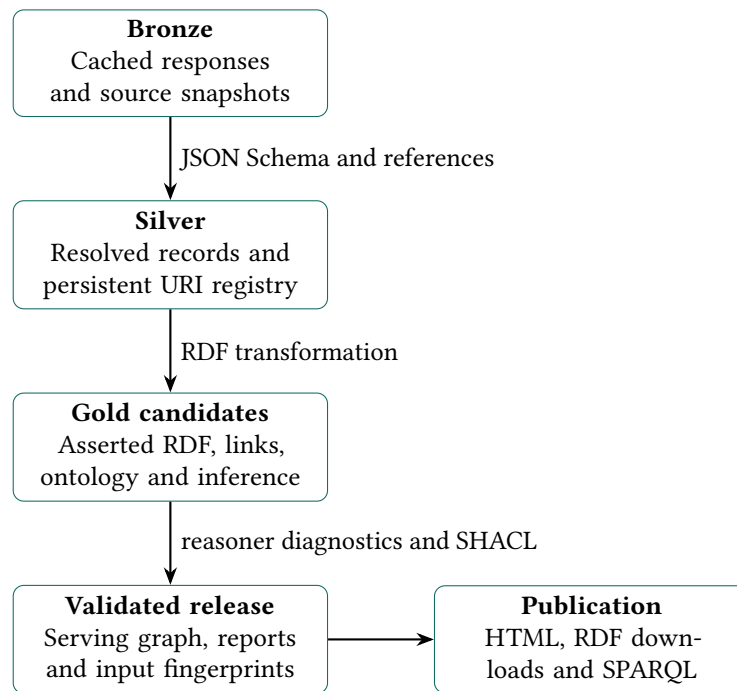


Figure 1: Release flow. A failed validation prevents replacement of validated inference and the serving graph; changed inputs invalidate the publication fingerprint.

selected instance-level entailments, and dataset metadata. The layers improve inspectability, but the implementation rewrites snapshot files during reconstruction. Bronze is therefore not an immutable append-only archive; Git revisions provide versioned snapshots.

Silver validation checks schema constraints and selected cross-field rules. The audit adds checks for future years, impossible calendar dates, missing cross-collection references, and invalid leadership references. These checks complement, rather than replace, semantic review of the extraction policy.

After RDF generation, literal ranges and functional-field cardinalities are validated before reasoning can merge object values through equality. The rule engine’s own error messages are retained in the consistency diagnostics. SHACL then checks the asserted and retained inferred graph. A failure stops the release step. On success, a validation record binds the checked release to its ontology, shapes, Silver files, component RDF files, and serving graph. The publisher rejects missing or stale validation records.

Fingerprints normalize text line endings to LF so that Git checkout on Windows and Linux does not create false stale-release failures. This is a content-integrity check, not a cryptographic signature or a guarantee against someone deliberately editing both the inputs and validation record. Reproduction of RDF is evaluated semantically; timestamps, blank-node labels, and serialization order can change without changing graph meaning.

4.2 Identity and URI persistence

A readable slug is used when a resource is first assigned a local URI. The revised minter consults the committed `uri_map.json` before allocating anything new. Renaming an entity or inserting another entity with the same label does not change a registered URI. Retired identifiers remain reserved. Source-key changes, such as a new QID replacing an article-based key, still require an explicit identity migration policy.

The previous person integration merged unidentified leaders by normalized personal name across all institutions. The audit found six names used across multiple institutions; splitting these records adds seven person records. This does not prove that seven new real people exist. It represents uncertainty

conservatively: records without a shared identifier are separated by institution until identity can be established. The former ambiguous URI is reserved but is not asserted as identical to either replacement.

4.3 Provenance and disputed values

Institution records use `prov:wasDerivedFrom` to reference Wikidata items and specific Wikipedia revisions. Conflict and fallback logs provide additional evidence for some chosen fields. This is entity-level provenance plus partial field-level logging; the project does not yet attach complete provenance to every triple. A Wikidata entity URI is also not a pinned statement revision.

The earliest reported foundation year is the current integration policy and may refer to a predecessor. Five records contain a different year in the infobox foundation date. The revised exporter retains such a date under `vnedu:reportedFoundingDate`, rather than asserting it as an unqualified `schema:foundingDate`. This preserves evidence without pretending that the historical ambiguity has been adjudicated. Distinguishing predecessor origin, legal establishment, and opening would require dated event records and authoritative supporting sources.

5 Ontology design and relationship semantics

5.1 Modules

The ontology has 34 named local classes, 33 local object properties, and 21 local datatype properties in the audited release. Table 1 summarizes the main relationship choices.

Table 1: Main modules and intended distinctions.

Module	Main entities or properties	Interpretation
Organizations	Institution, governing body, company; memberOf, branchOf, governedBy, ownedBy	Membership, branch status, governance, and ownership are separate relations.
Oversight	stateManagedBy	General state oversight is distinct from direct governance.
People	Person, InstitutionLeader, Alumnus; rector, director, councilChair	Leadership roles differ; education records do not certify graduation.
Geography	Province, FormerProvince, Region; locatedIn, partOf, mergedInto	Geographic crosswalks support aggregation under successor boundaries.
Education	AcademicProgram, Major, FieldOfStudy; offersProgram, ofMajor, inField	Example programs connect institutions to subjects; these are not an official offering inventory.
Identity	owl:sameAs, skos:closeMatch, foaf:isPrimaryTopicOf	Identity, approximate concept mapping, and document-topic links are not interchangeable.

Disjointness axioms separate selected organizational and geographic categories. Their usefulness depends on defensible category definitions. A statement that violates a disjointness axiom could indicate erroneous data or an overly rigid ontology, so a detected conflict should trigger inspection rather than automatic deletion of evidence.

Vocabulary alignments use subclass and subproperty relationships. Such alignments still carry logical implications; choosing a weaker alignment than equivalence does not make it semantically harmless. The current inference run does not import and validate the complete external ontologies. Its absence of detected conflicts therefore applies to the local reasoning scope, not every possible external combination.

5.2 Materialization

The system computes a rule closure over the local ontology and asserted data and retains selected new instance-level statements. External identity links are joined into the serving graph after this step. This avoids equality-driven copying to remote identifiers, but means the published union is not a complete closure including all owl:sameAs effects. The reported inferred-triple count describes the retained subset, not the full internal closure.

Representative rules are:

$$\begin{aligned} \text{locatedIn} \circ \text{partOf} &\sqsubseteq \text{locatedIn}, \\ \text{locatedIn} \circ \text{mergedInto} &\sqsubseteq \text{locatedIn}, \\ \text{governedBy} \circ \text{subordinateTo} &\sqsubseteq \text{governedBy}, \\ \text{offersProgram} \circ \text{ofMajor} &\sqsubseteq \text{trainsMajor}. \end{aligned}$$

For example, an institution assigned to a former province can also be related to the successor province and its larger region. Inverse properties expose the same relationship in the opposite direction without manually storing both directions in the asserted graph.

The model uses the owlrl implementation of OWL 2 RL/RDF rules. This is an implementation statement, not a formal certificate that the ontology belongs to the OWL 2 RL profile. In particular, temporal datatypes such as xsd:gYear and xsd:date require care relative to the profile's datatype restrictions. Conformance and completeness claims require a dedicated profile check and explicit semantics, beyond the present regression suite.

5.3 Geographic and temporal limitations

The bornIn chains are geographic containment crosswalks. Relating a birthplace to a province created in 2025 does not mean that province existed when the person was born. Similarly, historical institutions should not automatically be counted as currently operating just because their location can be mapped to a current province. Queries for current institutions should explicitly consider dissolution evidence and its incompleteness.

The pipeline lacks a general temporal model for leadership terms, student counts, legal type changes, and geographic versions. Absence of an end date is not proof of a current appointment. Birth-date collection currently loses source precision, and suppressing January 1 dates is only a heuristic: it can suppress real birthdays and cannot establish the precision of all other dates. These source-quality limitations remain documented work, not conclusions resolved by successful reasoning.

6 Linking and publication

6.1 External relations

The link graph contains 3,674 triples. Wikidata, ROR, GeoNames, and selected DBpedia links use owl:sameAs; subject mappings use skos:closeMatch; Wikipedia article links use foaf:isPrimaryTopicOf. The last of these relates an entity to a document and must not be counted as an identity assertion. Repeated attempts to add the same link no longer inflate linkset counters.

Shared identifiers are useful identity evidence, but errors and historical scope mismatches can occur in the source mappings. Some organizational links are discovered through label matching with contextual filters. These deserve separate manual review. The audit does not certify every identity link or every administrative boundary match.

6.2 Representations and deployment

The static publisher generates HTML pages, embedded JSON-LD, and RDF files. The dynamic server provides query handling and RDF content negotiation. The canonical namespace currently belongs to GitHub Pages; the dynamic service has another origin. Content negotiation at the dynamic service does not establish that the canonical static URI performs the same negotiation.

The implementation supports several practices associated with five-star Linked Open Data: open formats, HTTP identifiers, RDF descriptions, and outward links. This report treats that as a design assessment, not a certification. It does not reassert the previous report's live availability measurements. Operational claims need dated requests to the actual deployed revision. The repository declares data licensing and retains media metadata; this audit is not a legal verification of the license of every upstream asset.

The publisher now checks its output path before recursive replacement. It refuses the project root, external paths, symbolic links, and existing directories without generated-site markers. This prevents a misconfigured output environment variable from deleting source data or the project itself.

7 Query execution and security review

The original SPARQL policy searched for textual patterns containing whitespace after SERVICE and FROM. Valid forms without that whitespace bypassed the filter. The audit reproduced this locally using loopback targets without issuing an outbound request. The revised policy inspects parsed SPARQL operations and permits only two exact HTTPS service endpoints. It rejects dataset-loading clauses, variable service targets, alternate ports, credentials, and arbitrary paths. Security keywords inside string literals are not treated as operations. The local outbound HTTP handler also rejects redirects.

The previous timeout waited only for `Graph.query` to return. `RDFLib` can evaluate SELECT bindings lazily, so result evaluation could continue beyond the limit. Cancelling a future also cannot stop an already running Python thread. The revised local backend evaluates and serializes each query in a disposable process, terminates it on timeout, and limits concurrent workers. An expensive Cartesian-product regression test checks that capacity is released and a subsequent small query succeeds.

The local backend limits serialized responses to 8 MiB. This does not impose a hard memory limit during intermediate evaluation, and parsing the public request still occurs in the web process. A hostile-workload production deployment therefore needs process or container memory limits, request-rate controls, and appropriate upstream isolation. The process-per-query approach also reloads the graph and has a throughput cost. Earlier in-memory latency measurements do not describe this implementation.

The Fuseki proxy supplies an execution timeout parameter and does not follow proxy redirects. Actual query cancellation and remote SERVICE egress remain responsibilities of the configured Fuseki instance. They were not tested against a live Fuseki deployment in this audit. Raw POST requests with `application/sparql-query` are now accepted, alongside form-encoded query requests.

8 Evaluation and results

8.1 Release measurements

All values in Table 2 are generated by `audit/release_metrics.py` from the validated local release. That script also checks that the serving graph is isomorphic to the union of its declared components. It does not request data from a remote endpoint.

Table 2: Measured size of the corrected local release.

Quantity	Count
Educational-organization records	300
Classified higher-education institutions	271
Person records	1,702
Institution-leader records	221
People with recorded education relations	1,481
Current / former province-level units	34 / 29
Illustrative programs / majors	78 / 37
Asserted data triples	22,931
Retained inferred triples	21,018
Link-graph triples	3,674
Ontology / dataset-metadata triples	777 / 96
Distinct triples in serving union	48,492
Local resource IRIs appearing as subjects	2,248

In the corrected snapshot, selected SHACL checks report 0 violations, 34 warnings, and 1 informational result. Warnings include missing foundation years and institution locations. Warning-tolerant conformance means the configured release policy permits these omissions; it does not mean all records are complete. The consistency diagnostics report no detected local violations after the fixes.

Field coverage uses all 300 institution records as its denominator: 278 have a selected foundation year, 288 a province reference, 256 an ownership value, 237 a website, 175 coordinates, 60 an admission code, and 28 a student-count value. These are presence measurements, not estimates of truth, timeliness, or national coverage. In particular, 271 higher-education classifications and 300 educational-organization records are different denominators and should not be interchanged.

8.2 Link experiment: agreement, not independent accuracy

The available experiment uses 86 institutions with English labels and identifier-derived DBpedia reference targets. A local token-Jaccard matcher selects the highest-scoring target from that closed reference candidate set. Table 3 reports agreement with the reference. It is not a run of the Silk software.

Table 3: Token-Jaccard agreement with the identifier-derived reference.

Threshold	Predictions	Disagreements	Ref. precision	Ref. recall	Ref. F1
0.2	85	13	84.7%	83.7%	0.842
0.4	80	8	90.0%	83.7%	0.867
0.6	76	6	92.1%	81.4%	0.864
0.8	68	4	94.1%	74.4%	0.831
1.0	58	1	98.3%	66.3%	0.792

Let M be the set of predictions, R the reference pairs, and $TP = |M \cap R|$. The reported precision is $TP/|M|$, recall is $TP/|R|$, and F1 is their harmonic mean. Calling R independent ground truth would be incorrect because the identifier linker generated it. Scoring that same linker against R yields perfect agreement by construction. The previous claim of 100% linking accuracy is therefore withdrawn.

The selected sample excludes unlinked institutions and targets outside the reference set. These restrictions affect task difficulty and coverage. The threshold with the highest F1 on this sample is a descriptive observation, not a validated threshold chosen on a separate development set. A defensible

accuracy study needs independently annotated links and non-links, dated evidence, an adjudication procedure, and a held-out test set.

8.3 Regression evidence

The audit adds counterexamples for compact SPARQL syntax, exact service endpoint policy, keywords inside literals, raw query POSTs, cancellation and worker recovery, datatype diagnostics, literal ranges, publication gates, stale release records, persistent URIs, same-name identity collisions, offline cache misses, impossible dates, and dangerous publisher paths. Eight initial counterexamples failed before correction and passed after the relevant fixes. The final test-run result and command are recorded in the companion audit notes rather than a manually maintained badge.

The full pipeline is reconstructed from the shipped cache, and both static-site variants are built before the server integration suite. A successful run establishes that the tested release follows the intended local contracts. It does not constitute a penetration test, an independent factual annotation study, or a load test of the public deployment.

8.4 Treatment of earlier comparisons

The earlier report's 21-fact comparison with a predecessor repository is a small, hand-selected demonstration. Its benchmark script relies on a machine-specific path and permissive answer matching, and the underlying cases do not have a complete independent annotation protocol. Its scores are not used here as an accuracy estimate. Claims about 349 predecessor contradictions, graph-density superiority, live link availability, and public-endpoint latency are likewise not repeated as newly verified findings.

Graph density is especially sensitive to which nodes and predicates are counted. Adding external nodes or inferred inverse edges changes the measurement without necessarily adding useful or correct information. A future comparison should define a common node universe, edge projection, deduplication policy, and frozen input revisions before reporting degree statistics.

9 Threats to validity and remaining work

Construct validity. Logical consistency, SHACL conformance, field presence, and link agreement measure different things. None alone measures real-world correctness. Program examples and name-based legal-type classification further restrict what answers can be interpreted as authoritative.

Internal validity. Source precedence can select a plausible but incorrect value. Earliest-year selection conflates events unless the event meaning is modeled explicitly. Some unqualified dates and organizational relationships need source-level adjudication. The revised person split prevents unsupported merges, but may also separate records belonging to the same real person; this is an explicit uncertainty trade-off.

External validity. Coverage favors institutions and people documented in the selected community sources. The 86-link experiment omits open-world candidates and cannot be generalized to all Vietnamese institutions. National completeness requires comparison with a dated authoritative institutional register.

Operational validity. Local tests do not prove public availability, deployed revision identity, traffic capacity, or Fuseki egress behavior. Memory limits and endpoint rate controls remain deployment concerns. Snapshot timestamps record reconstruction or collection events as implemented; they must not automatically be interpreted as the dates on which the underlying facts became true.

The next substantive data-model improvements are dated leadership appointments, source-qualified measurements, event-specific foundation histories, and assertion-level provenance. A subsequent research evaluation should use a preregistered sampling method, an independently reviewed reference set, and a clear distinction between discovery coverage, linking coverage, and correctness. Those additions require evidence not present in the current repository and are not supplied by inventing dates, labels, or outcomes.

10 Conclusion

VN-Edu LOD demonstrates a working integration and publication pipeline for a Vietnamese education knowledge graph. The audited implementation strengthens the boundary between untrusted queries and execution, between candidate data and publication, and between source labels and entity identity. Its measured result is a locally validated release with 48,492 distinct serving triples and explicit limitations on factual, temporal, and operational interpretation. The available evidence supports the system as an educational Linked Data artifact; it does not establish perfect accuracy or complete coverage of Vietnamese higher education.

A Reproducing the release and report

Use Python 3.12 and the pinned dependencies. After installation, the default pipeline uses the shipped cache and fails on missing cached responses. Run from the repository root:

```
python -m pip install -r requirements.txt
python run_all.py --no-install
python audit/release_metrics.py
```

Build the dynamic-server UI before running the complete integration suite. In PowerShell:

```
$env:VNEDU_SITE_DIR = 'site_server'
$env:VNEDU_SITE_ROOT = '/'
python scripts/step7_publish.py
python -m pytest tests -q
```

Unset the two UI environment variables before building the GitHub Pages variant. The default pipeline already builds that variant. Existing output directories must have the publisher's generated-site markers.

From docs/report, compile the English edition with an installed TeX distribution:

```
latexmk -lualatex -interaction=nonstopmode -halt-on-error main-en.tex
```

The PDF uses Libertinus and Noto Sans Mono. The numeric macros come from tables/audit-metrics.tex; regenerate them after data or ontology changes. data/reports/audit-metrics.json contains the corresponding machine-readable values. The original Vietnamese source and PDF remain available as historical project artifacts.

B A competency-query example

This query counts classified institutions by current province using the retained geographic inference. It excludes institutions explicitly marked as defunct; the absence of that marker still does not prove current operation.

```
PREFIX vnedu: <https://pham-ng.github.io/Vietnam-University-Knowledge-Graph-ver2/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?province (COUNT(DISTINCT ?institution) AS ?count)
WHERE {
  ?institution a vnedu:HigherEducationInstitution ;
               vnedu:locatedIn ?p .
  ?p a vnedu:Province ; rdfs:label ?province .
  FILTER(LANG(?province) = "vi")
  FILTER NOT EXISTS { ?institution a vnedu:DefunctInstitution }
}
GROUP BY ?province
ORDER BY DESC(?count)
```

The example illustrates an essential reporting distinction: a query can execute correctly against the model while its answer remains dependent on the completeness and meaning of the underlying evidence.

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